

Synthetic Mortgage Underwriting Document

Document ID: ANX-MTG-0001
Document Type: Synthetic Mortgage Underwriting Case File
Status: Training / Testing Only
Borrower: Jonathan Michael Carter
Property: 1847 Willow Creek Drive, Austin, TX 78745
Loan Purpose: Primary Residence Purchase
All information in this document is fictional.

1. LOAN APPLICATION SUMMARY

1.1 Borrower Information

Primary Borrower: Jonathan Michael Carter
Date of Birth: 14 March 1988
Marital Status: Married
Dependents: 2
Citizenship: U.S. Citizen

Current Address: 1847 Willow Creek Drive, Austin, TX 78745

Phone: (512) 555-0187
Email: jonathan.carter@example.test

Synthetic Data Notice: All names, addresses, contact information, account numbers, and financial figures in this document are fictional and created exclusively for software testing.

1.2 Loan Request

Field	Value
Loan Type	Conventional Fixed-Rate Mortgage
Loan Purpose	Purchase
Occupancy	Primary Residence
Purchase Price	\$400,000
Requested Loan Amount	\$320,000
Down Payment	\$80,000
Proposed Interest Rate	6.500%
Loan Term	30 years
Property Type	Single-Family Residence
Number of Units	1

1.3 Application Identifier

Application ID: ANX-2026-0001
Application Date: 10 September 2026
Underwriting Status: Pending Final Decision

2. EMPLOYMENT AND INCOME

2.1 Current Employment

Employer: Lone Star Technology Solutions, Inc.
Position: Senior Software Engineer
Employment Type: Full-Time
Employment Start Date: 7 June 2019
Employment Status: Active

2.2 Base Compensation

Annual base salary: **\$120,000**

Monthly gross base income: **\$10,000**

2.3 Variable Compensation

The borrower received the following documented annual bonuses:

Year	Bonus
2023	\$10,000
2024	\$12,000
2025	\$14,000

Three-year average bonus: **\$12,000 per year**

Monthly qualifying bonus income: **\$1,000**

2.4 Qualifying Income

Income Component	Monthly Amount
Base Salary	\$10,000
Average Bonus	\$1,000
Other Qualifying Income	\$0
Total Gross Monthly Income	\$11,000

2.5 Employment Verification

Employment was verified through a fictional automated employment verification service on:

10 September 2026

Verification result:

Employment confirmed — active

3. CREDIT PROFILE

3.1 Credit Score

Representative FICO Score: 760

The credit report indicates a strong credit history with no reported bankruptcy, foreclosure, or short sale within the underwriting review period.

3.2 Monthly Liabilities

Liability	Monthly Payment
Auto Loan	\$550
Student Loan	\$300
Credit Card Minimum Payments	\$150
Personal Loan	\$200
Proposed Mortgage P&I	\$2,022
Property Taxes	\$450
Homeowners Insurance	\$150
HOA	\$100
Total Monthly Obligations	\$3,922

3.3 Debt-to-Income Calculation

Gross monthly qualifying income:

\$11,000

Total monthly obligations:

\$3,922

DTI:

$\$3,922 \div \$11,000 \times 100 = 35.65\%$

Calculated DTI: 35.65%

3.4 Credit Findings

- No bankruptcy identified.
- No foreclosure identified.
- No recent short sale identified.
- Representative FICO score is 760.
- Revolving credit utilization is reported at 18%.
- Payment history shows no 30-day late payments during the reviewed period.

4. ASSETS AND RESERVES

4.1 Verified Assets

The borrower provided statements for the following fictional accounts.

Asset	Verified Balance
Checking Account	\$38,500
Savings Account	\$52,000
Brokerage Account	\$46,500
Retirement Account	\$85,000
Total Reported Assets	\$222,000

4.2 Funds Required for Closing

Requirement	Amount
Down Payment	\$80,000
Estimated Closing Costs	\$9,500
Prepaid Taxes and Insurance	\$3,500
Required Cash to Close	\$93,000

4.3 Remaining Liquid Assets

Total reported assets:

\$222,000

Estimated cash required to close:

\$93,000

Remaining assets:

\$129,000

The borrower therefore demonstrates sufficient documented assets for the stated cash-to-close requirement.

4.4 Reserve Calculation

Proposed total monthly housing obligation: **\$2,722**

Remaining liquid assets: **\$129,000**

Approximate reserves: **47.39 months**

5. PROPERTY INFORMATION

5.1 Subject Property

Property Address: 1847 Willow Creek Drive, Austin, TX 78745

Property Type: Single-Family Residence

Year Built: 2018

Living Area: 2,150 square feet

Bedrooms: 4

Bathrooms: 3

Occupancy: Primary Residence

5.2 Purchase Information

Contract Purchase Price: \$400,000

Borrower Down Payment: \$80,000

Requested Mortgage: \$320,000

5.3 Appraised Value

The fictional appraisal reports:

Appraised Value: \$410,000

The appraised value exceeds the contract purchase price by: **\$10,000**

5.4 Loan-to-Value Calculation

Requested loan amount: **\$320,000**

For this test case, the underwriting system uses the lower of purchase price or appraised value for the standard purchase LTV calculation.

Value used: **\$400,000**

LTV:

$\$320,000 \div \$400,000 \times 100 = 80.00\%$

Calculated LTV: 80.00%

6. PROPERTY COMPARABLES

6.1 Comparable Sales

The fictional appraisal contains the following comparable properties.

Comparable	Sale Price	Distance	Size
Comparable 1	\$405,000	0.4 miles	2,100 sq ft
Comparable 2	\$395,000	0.7 miles	2,180 sq ft
Comparable 3	\$415,000	0.9 miles	2,220 sq ft

Average comparable sale price:

\$405,000

Subject property appraised value:

\$410,000

6.2 Property Valuation Finding

The appraised value of **\$410,000** is considered reasonably supported by the supplied comparable sales.

No material property condition issue affecting value was identified in the fictional appraisal.

6.3 Property Risk Flags

- No flood-zone issue identified in the supplied report.
- No major structural defect identified.
- No unresolved safety issue identified.
- Property is classified as a single-family residence.
- Intended occupancy is primary residence.

7. DOCUMENT CHECKLIST

7.1 Documents Received

The following documents were supplied for underwriting:

Document	Status
Uniform Residential Loan Application	Received
Government-Issued Identification	Received
Two Years of W-2 Forms	Received
Two Years of Tax Returns	Received
Recent Pay Stubs	Received
Employment Verification	Received
Bank Statements	Received
Brokerage Statement	Received
Retirement Account Statement	Received
Credit Report	Received
Property Appraisal	Received
Purchase Contract	Received
Homeowners Insurance Quote	Received

7.2 Missing Documents

No critical underwriting document is currently identified as missing.

7.3 Documentation Consistency

The supplied income documentation is consistent with the employment information.

The supplied asset documentation is consistent with the stated funds required for closing.

The credit report is consistent with the liabilities used in the DTI calculation.

8. UNDERWRITING GUIDELINES

8.1 Synthetic Qualification Criteria

For this test case, the underwriting engine should evaluate the following fictional criteria.

Criterion	Test Threshold
Minimum FICO	700
Maximum DTI	43%
Maximum Standard LTV	80%
Minimum Reserves	6 months
Critical Missing Documents	0
Bankruptcy in Review Period	Not permitted
Foreclosure in Review Period	Not permitted

8.2 Borrower Comparison

Criterion	Borrower	Requirement	Result
FICO	760	≥ 700	Pass
DTI	35.65%	$\leq 43\%$	Pass
LTV	80.00%	$\leq 80\%$	Pass
Reserves	47.39 months	≥ 6 months	Pass
Critical Missing Documents	0	0	Pass
Bankruptcy	None	None	Pass
Foreclosure	None	None	Pass

8.3 Initial Underwriting Assessment

Based solely on the synthetic information supplied in this case file:

The borrower satisfies all defined test-case qualification criteria.

The case is therefore suitable for testing an **Approve** recommendation.

9. REGULATORY AND COMPLIANCE TEST DATA

9.1 Compliance Review

The fictional compliance process checks the following areas:

RESPA

The underwriting workflow must preserve required disclosures and must not introduce prohibited settlement-service practices.

TILA

The system must ensure that applicable loan terms, costs, and disclosures are handled according to applicable disclosure requirements.

Fair Housing

The decision must be based on relevant financial, credit, property, and underwriting criteria.

Protected characteristics must not be used as decision factors.

9.2 Fair Lending Test

The following information must **not** influence the underwriting recommendation:

- Race
- Color
- Religion
- Sex
- National origin
- Familial status
- Disability

The fictional underwriting engine must base the recommendation on documented qualification criteria.

9.3 Adverse Action Scenario

If a future test case fails one or more qualification criteria, the system should:

1. Identify the failed criterion.
2. Preserve supporting evidence.
3. Avoid unsupported explanations.
4. Generate an appropriate adverse-action reason.
5. Avoid using protected characteristics as a reason for the decision.

10. DECISION AGENT TEST INPUT

10.1 Consolidated Risk Profile

Borrower: Jonathan Michael Carter, **Gross Monthly Income:** \$11,000, **Monthly Debt Obligations:** \$3,922, **DTI:** 35.65%, **FICO:** 760, **Purchase Price:** \$400,000, **Appraised Value:** \$410,000, **Loan Amount:** \$320,000, **LTV:** 80.00%, **Liquid Assets After Closing:** \$129,000, **Reserves:** 47.39 months, **Critical Missing Documents:** 0, **Bankruptcy:** None, **Foreclosure:** None

10.2 Expected Decision

Decision: APPROVE

Risk Level: LOW

10.3 Expected Decision Reasoning

The fictional borrower meets all defined test-case underwriting thresholds:

- FICO exceeds the minimum requirement.
- DTI is below the maximum permitted threshold.
- LTV is at the maximum permitted standard threshold but does not exceed it.
- Documented reserves substantially exceed the minimum requirement.
- No critical underwriting document is missing.
- No bankruptcy or foreclosure is identified in the review period.
- Property valuation is supported by the supplied comparable sales.

10.4 Decision-Agent Output Requirements

A compliant Decision Agent should return:

Decision: Approve

Risk Level: Low

Key Factors:

1. FICO = 760.
2. DTI = 35.65%.
3. LTV = 80.00%.
4. Reserves = 47.39 months.
5. No critical missing documents.
6. No bankruptcy or foreclosure identified.

Evidence: Every material conclusion must reference the corresponding document page.

11. SYNTHETIC EDGE-CASE DATA

This section exists specifically to test retrieval and reasoning.

11.1 Boundary Conditions

The loan has:

LTV = 80.00%

Therefore:

- LTV is **not above** the 80% limit.
- LTV should be treated as **passing** under the supplied test criteria.

11.2 DTI Boundary

The maximum permitted DTI is: **43%**

Borrower's DTI: **35.65%**

DTI therefore passes with a margin of: **7.35 percentage points**

11.3 Asset Boundary

Required cash to close:

\$93,000

Remaining liquid assets:

\$129,000

The borrower retains sufficient funds after estimated closing requirements.

11.4 Potentially Confusing Values

The property has two relevant values:

Contract Purchase Price: \$400,000

Appraised Value: \$410,000

For this test case's standard purchase LTV calculation, the system uses the **lower value of \$400,000**.

Therefore:

\$320,000 / \$400,000 = 80.00%

12. DOCUMENT METADATA AND TESTING NOTES

12.1 Document Metadata

Document ID: ANX-MTG-0001

Version: 1.0

Created For: ANexus RAG Testing

Data Classification: Synthetic / Non-Production

Contains Real PII: No

Contains Synthetic PII Patterns: Yes

12.2 Intended RAG Tests

This document is intended to test:

- PDF text extraction
- Page-level metadata
- OCR fallback
- Chunking
- Dense retrieval
- BM25 retrieval
- Hybrid retrieval
- Cross-encoder reranking
- Page citations
- PII detection
- PII redaction
- Multi-turn questions
- Numerical reasoning
- Evidence-grounded answers
- Missing-information handling
- Decision-agent reasoning
- Compliance-related retrieval
- Boundary-value reasoning

12.3 Important Testing Rule

The assistant must **not invent information that is absent from this document.**

If asked for information that is not contained in the uploaded document, ANexus should respond that the information is not available in the uploaded document.